ASIFA SIRAJ

Full Stack Developer | React.js, Node.js & MySQL

as1615533@gmail.com | 03128571654 | Karachi, Pakistan | personal-portfolio | linkedin.com/in/asifa-siraj | github.com/AsifaSiraj

PROFILE

Computer Science and IT graduate from NED University of Engineering & Technology with a foundation in full-stack web development using JavaScript, React.js, Node.js, Express.js, and MySQL. Experienced in developing responsive web interfaces, building database-backed applications, integrating REST APIs, and debugging existing software. Projects include a full-stack Library Management System and a hospital recommendation chatbot. Also bring experience in accessibility testing, software quality assurance, and Python-based AI applications. Seeking a Full Stack Developer Internship to contribute to real-world web applications and strengthen backend development skills.

EXPERIENCE

NED University of Engineering & Technology, INSPIRE Training Program, *Digital IC Design & Verification Trainee* 05/2026 – 09/2026

- Undergoing intensive training in Digital IC Design, RTL Design and Functional Verification using Verilog/SystemVerilog and industry-standard EDA tools.
- Designed and verified a RISC-V SoC with ECC-Protected Memory, including CPU datapath, instruction/data memory, ECC encoding/decoding and fault-injection mechanisms.
- Developed UVM-based verification environments with sequences, drivers, monitors, scoreboards, assertions and functional coverage.
- Worked with QuestaSim, ModelSim and Intel Quartus Prime for simulation, debugging, waveform analysis and FPGA-oriented development.
- Gained practical exposure to RISC-V architecture, AMBA protocols, assertions, constrained-random verification and RTL debugging.

Deserts Digital OÜ, *Digital Accessibility Intern* 07/2025 – 12/2025 | Remote
Internship

- Led WCAG 2.1 AA audits across web pages and PDFs using a structured QA/testing approach, identifying and remediating accessibility issues to improve screen-reader compatibility.
- Improved accessibility compliance scores by 90% by remediating contrast, ARIA labeling, and keyboard-navigation issues.
- Supported outreach and business-development coordination across assigned regions, contributing directly to deal-closing workflows for multiple client accounts.

DesertBIM, *Virtual Office Assistant* 08/2025 – 12/2025 | Remote
Freelance | Part-time, concurrent with Accessibility internship

- Provided administrative and organizational support, documentation, research, and cross-team coordination, alongside frontend development work.
- Coordinated calendars and communications to streamline day-to-day business processes for the team.

DIL Service Portal, NED University, *Frontend Web Developer* 01/2025 – 03/2025 | Hybrid

- Owned end-to-end delivery of responsive web interfaces for the DIL Service Portal using React and modern CSS, supporting 100+ active users.
- Analyzed and maintained existing portal components, improving code maintainability and fixing reported bugs (QA/debugging) to keep the application stable and testable.

Interns School, *Frontend Developer* 12/2024 – 01/2025 | Remote
Internship

- Developed and tested a COVID-19 testing web application, strengthening responsive design, JavaScript UI, and functional testing skills.
- Built and customized a personal portfolio site in WordPress, demonstrating CMS theme customization and frontend delivery.

Karachi Development Authority (KDA), *IT Support Staff* 03/2025 – 04/2025 | On-site
Internship

- Completed a structured orientation program covering KDA's IT department structure, system administration practices, and software maintenance protocols.
- Studied departmental workflows for web development, digital record-keeping, and IT infrastructure management within a public-sector environment.
- Attended technical briefings on KDA's e-governance platforms and internal software systems, gaining conceptual understanding of how digital services are delivered across departments.
- Developed a strong theoretical foundation in public-sector IT operations, motivating interest in returning for a hands-on trainee role to apply and build on this knowledge practically

TECHNICAL SKILLS

Programming Languages
JavaScript (ES6+), TypeScript, Python, PHP, HTML5, CSS3

Database & Data
SQL, MySQL, Relational Database Design, Database Management, Data Analysis Basics

Software Development
React.js, Node.js, Express.js, REST APIs, Next.js, Tailwind CSS, Bootstrap, WordPress

AI & Automation
LLM, AI Application Development, NLP

Testing & QA
SEO, Functional/Software Testing, UVM, System Verilog Assertions, Functional Coverage, Accessibility Testing, Functional Testing, Debugging

Digital IC / Verification
RTL Design, Verilog, SystemVerilog, RISC-V, FPGA, AMBA, ECC, UVM

Tools & Platforms
Git, GitHub, Microsoft Office, VS Code, QuestaSim, ModelSim, Intel Quartus Prime

Software Practices
Scalable & Testable Code, Application Maintenance, API Integration, Responsive Web Development, Agile/Scrum Basics

Accessibility
WCAG 2.1, ARIA, Screen Reader Testing, PDF Accessibility

KEY PROJECTS

Pointrix: Bus Management & Real-Time Tracking, *Dart, Flutter*
Final-year project delivering real-time transportation monitoring, giving users efficient access to bus and route information.

ReqMind-AI: Requirement Analysis System, *Python, LLM*
Built for the HEC Generative AI Hackathon; uses LLMs to extract functional and non-functional requirements from raw project input.

Noor: Islamic Web App, *TypeScript*
Built a full-featured Islamic companion app: prayer times, Qibla direction, Quran reader, Zikr tracking, and smart schedule planning.

Library Management System, *React, Express, MySQL*
Built a full-stack library system with book search, lending, returns, and admin reporting, using a scalable REST API backend, a MySQL relational database, and a React frontend.

Hospital Recommendation Chatbot, *Python, React, NLTK*
Built an NLP-based chatbot recommending hospitals by specialization, location, and budget, with voice input and text-to-speech.

RISC-V SoC with ECC-Protected Memory, *System Verilog*
Developed and verified a 32-bit RISC-V SoC with ECC-protected memory using SystemVerilog, UVM, and QuestaSim, including fault injection and memory/peripheral verification.

EDUCATION

BS Computer Science & IT, *NED University of Engineering & Technology* 09/2022 – 09/2026

Intermediate (Pre-Engineering), *Sir Syed Govt. Girls College* 07/2020 – 04/2022
Grade: A+

Matriculation (Computer Science), *High Star Public Sec School* 03/2015 – 03/2020
Grade: A+

CERTIFICATIONS

Generative AI App Developer: HEC Pakistan (2025)

Google Analytics: Skillshop (2025)

Basic Angular Development: NED University (2025)

Web Development: Coursera (2023)

HTML, CSS & JS for Web: Coursera (2023)